Staking Pools Analysis (Million ADA Units)

Dataset Filtering

Metric	Value
Pools in source	2941
Excluded not active pools ($active_stake = 0$)	224
Pools used in calculations (active only)	2717

Table 1: Filtering applied before running all calculations.

Global Stake and Pledge Summary (Million ADA)

Metric	Value
Total active stake	21568.605M
Total active pledge	2931.539M
Average active stake	7.938M
Average active pledge	1.079M

Table 2: Aggregate and average values in million ADA (active pools only).

Saturation, Zero Pledge, and Stake=Pledge (active pools only)

Metric	Count	Percentage
Saturation ≥ 1 (100%)	7	0.2576%
Active pledge = 0	224	8.2444%
Active stake = active pledge	329	12.1089%

Table 3: Counts and percentages among active pools only.

Distribution for stake = pledge subgroup

Margin distribution (12 groups)

Fixed cost distribution (ADA groups)

$pledge \times L$ Calculations (Million ADA, active pools only)

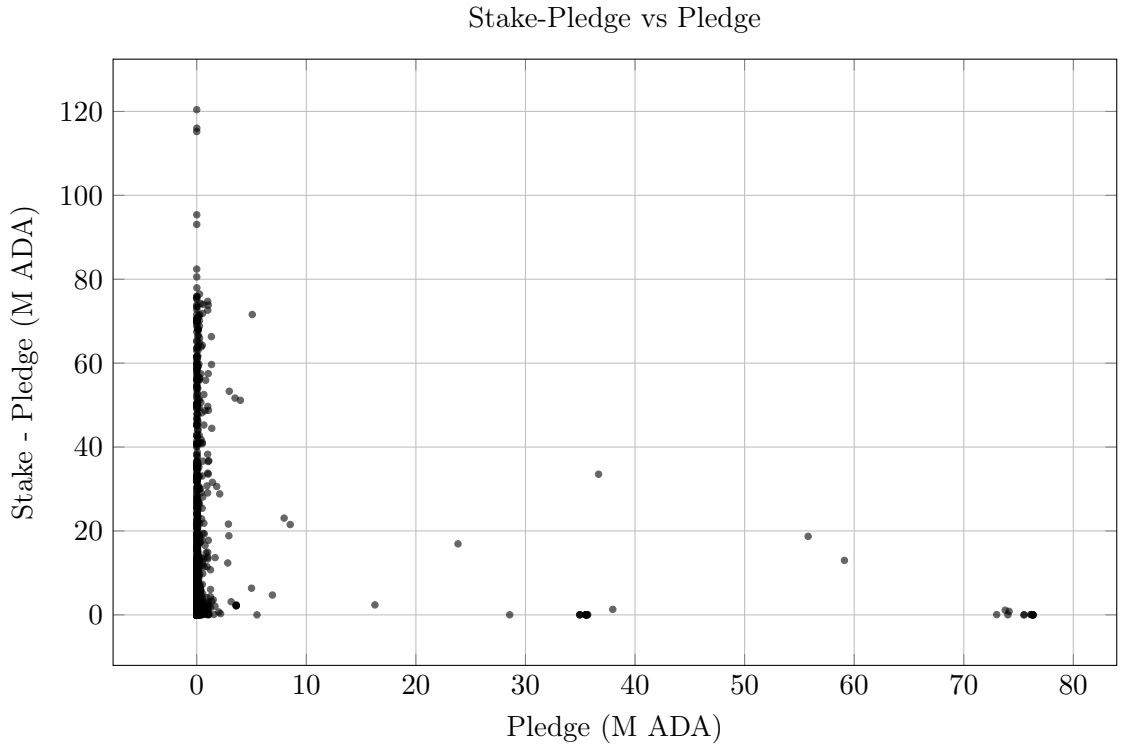
Margin group	Count	% of subgroup
=0	49	14.89%
(0.0,0.1]	223	67.78%
(0.1,0.2]	22	6.69%
(0.2,0.3]	1	0.30%
(0.3,0.4]	0	0.00%
(0.4,0.5]	0	0.00%
(0.5,0.6]	0	0.00%
(0.6,0.7]	0	0.00%
(0.7,0.8]	0	0.00%
(0.8,0.9]	1	0.30%
(0.9,1)	3	0.91%
=1	30	9.12%

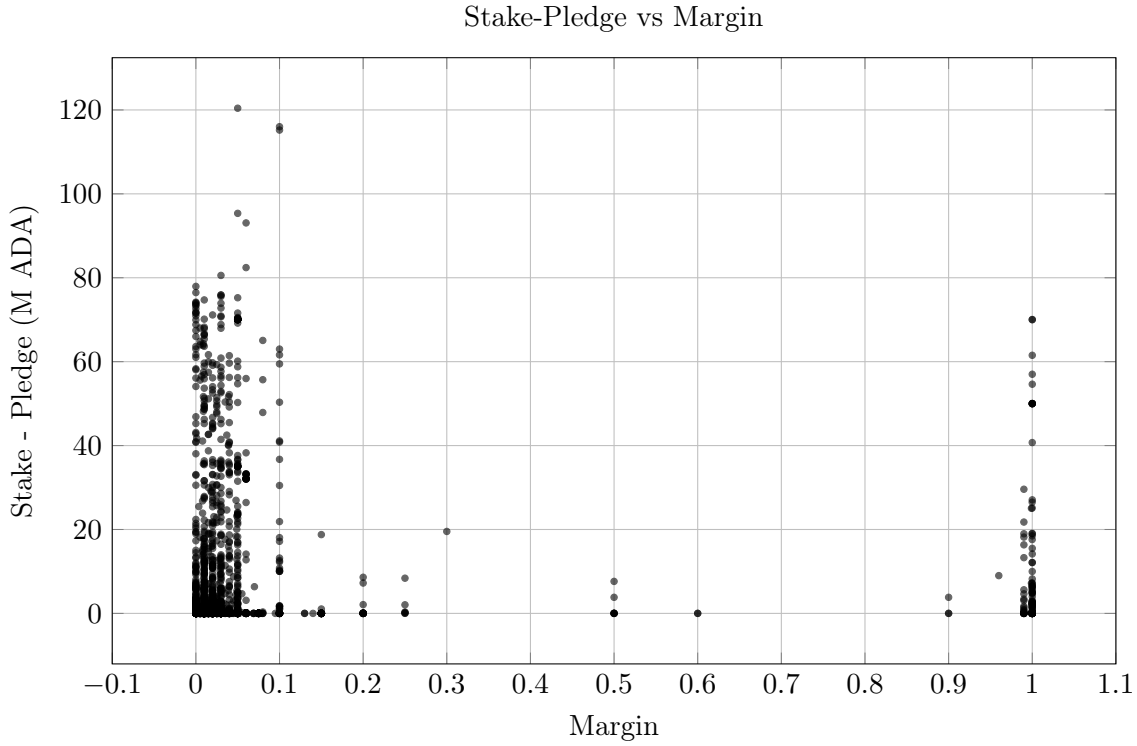
Fixed cost group [ADA]	Count	% of subgroup
=0	0	0.00%
(0,340)	31	9.42%
=340	218	66.26%
340-500	79	24.01%
>500	1	0.30%

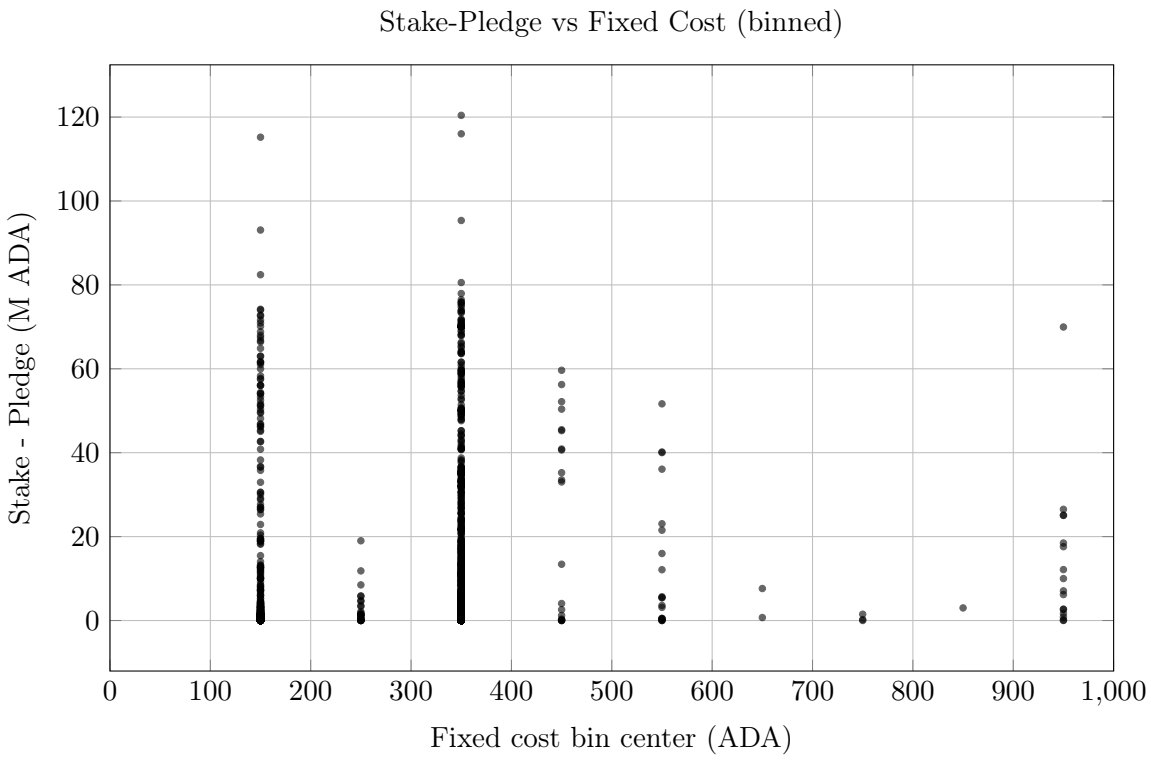
L	$\#(stake > pledge \cdot L)$	$\sum(stake - pledge \cdot L)$ [M ADA]	$\sum(stake - pledge \cdot L \mid pledge = 0)$ [M ADA]	$\#(stake < pledge \cdot L)$	$\sum(pledge \cdot L - stake)$ [M ADA]
1	2339	18726.811M	2747.535M	49	89.745M
10	1511	17401.376M	2747.535M	1206	25148.164M
25	1289	16195.18M	2747.535M	1428	67915.058M
100	998	13473.519M	2747.535M	1719	285058.847M

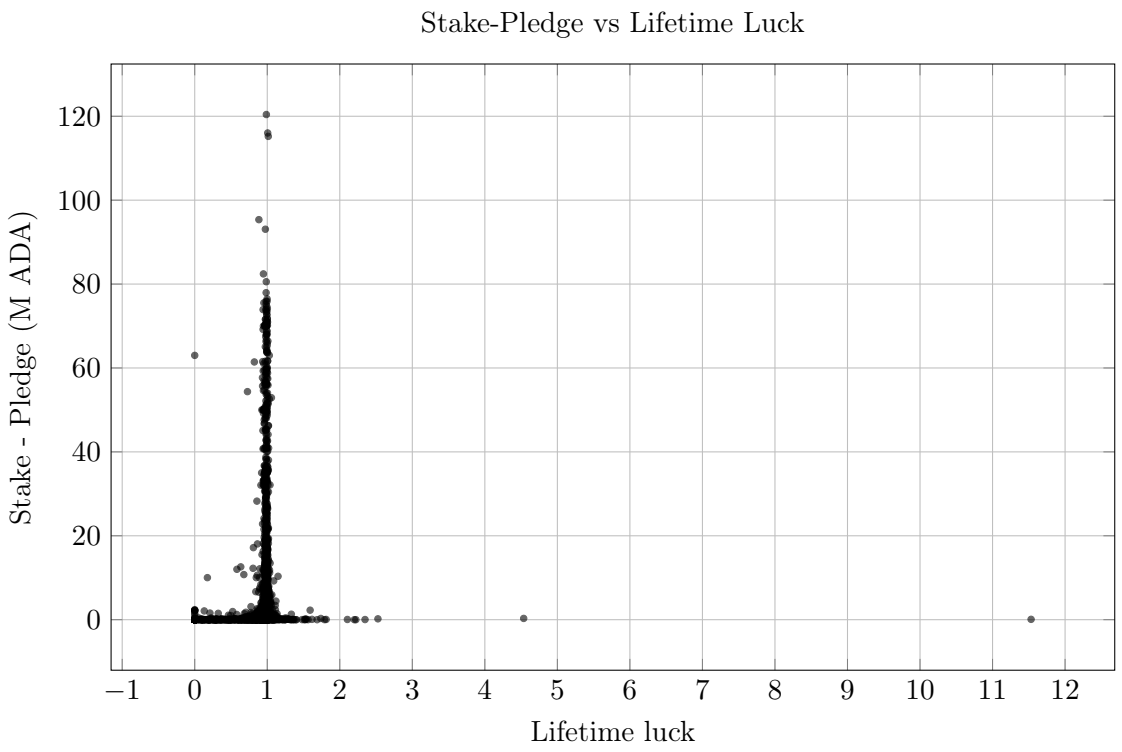
Table 4: Aggregate excess/deficit values in million ADA for each multiplier L (active pools only).

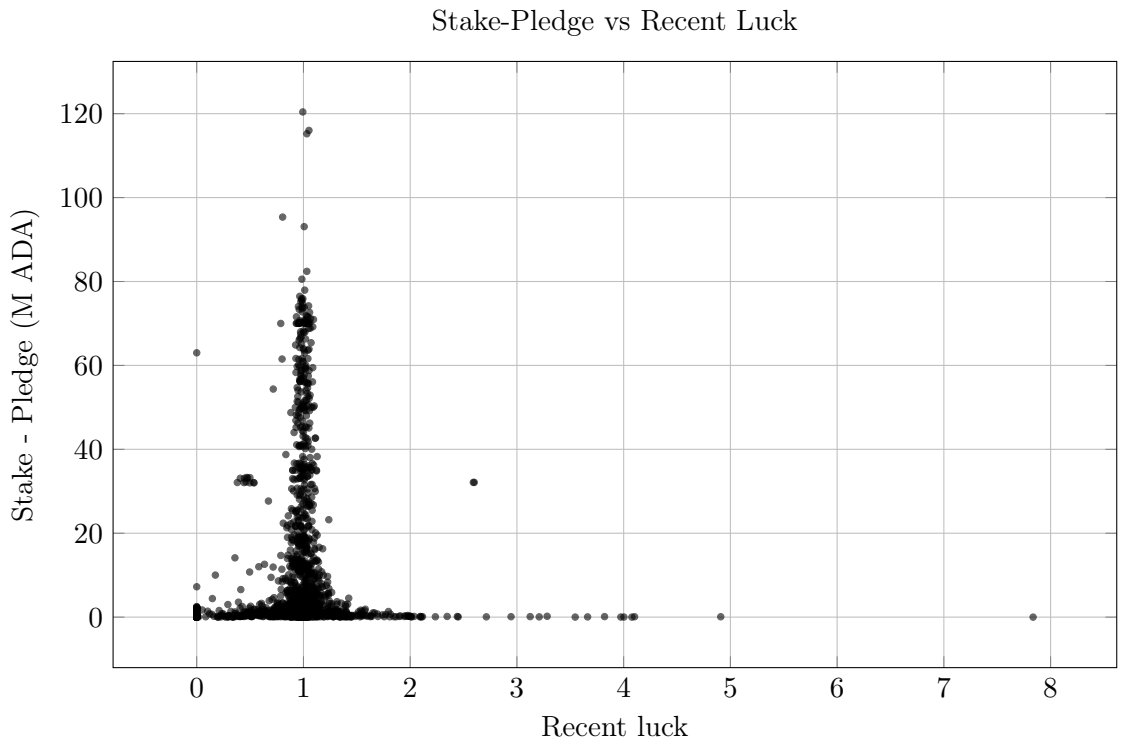
Charts for pools with stake > pledge

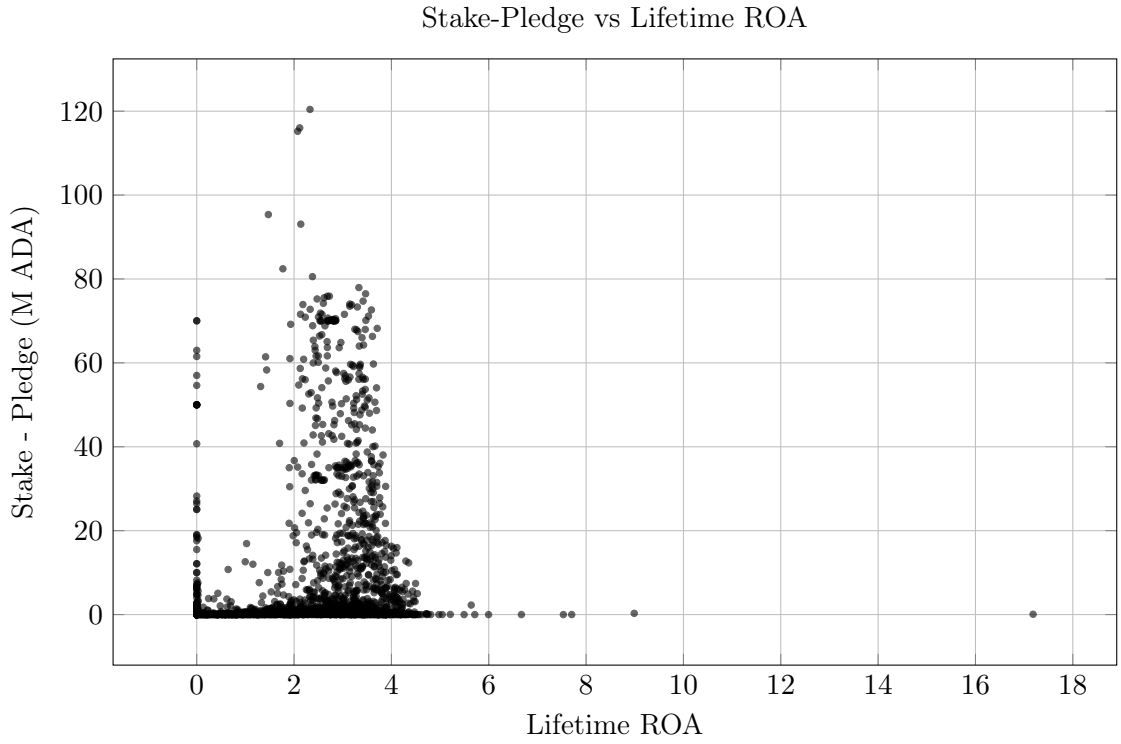


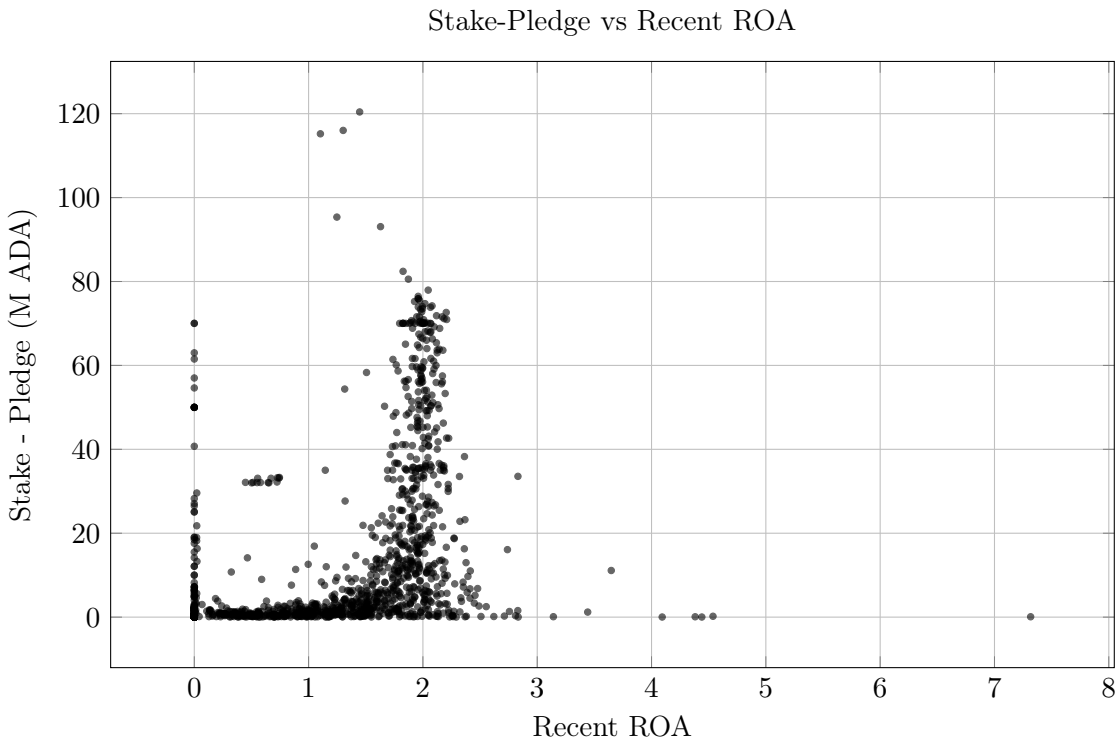



Stake-Pledge vs Fixed Cost (binned)
Stake - Pledge (M ADA)
120
100
80
60
40
20
0
0
100
200
300
400
500
600
700
800
900
1,000
Fixed cost bin center (ADA)


Stake-Pledge vs Lifetime Luck
Stake - Pledge (M ADA)
120
100
80
60
40
20
0
−1
0
1
2
3
4
5
6
7
8
9
10
11
12
Lifetime luck


Stake-Pledge vs Recent Luck
Stake - Pledge (M ADA)
120
100
80
60
40
20
0
0
1
2
3
4
5
6
7
8
Recent luck


Stake-Pledge vs Lifetime ROA
Stake - Pledge (M ADA)
120
100
80
60
40
20
0
0
2
4
6
8
10
12
14
16
18
Lifetime ROA



Multivariate Regression Results

Sample: **2339** pools with *active_stake* > *pledge*.
Dependent variable: `log(1 + (active_stake - pledge) in ADA)`

Model	ROA term(s)	R^2	Adj. R^2
A	lifetime_roa	0.6664	0.6656
B	recent_roa	0.6930	0.6922
C	lifetime_roa + recent_roa	0.7015	0.7006

Variable	Beta	Robust SE (HC1)	p-value
Intercept	5.1656	0.1467	0.0000
log_pledge	0.1664	0.0182	0.0000
margin	1.7314	0.3763	0.0000
fixed_cost_ada	0.0000	0.0000	0.0002
saturation	5.5399	0.3864	0.0000
log_delegators	0.9535	0.0434	0.0000
lifetime_roa	0.5655	0.0592	0.0000

Model A

Model B

Model C

Variable	Beta	Robust SE (HC1)	p-value
Intercept	5.8093	0.1498	0.0000
log_pledge	0.1304	0.0178	0.0000
margin	2.3094	0.3770	0.0000
fixed_cost_ada	0.0000	0.0000	0.0005
saturation	3.4660	0.3988	0.0000
log_delegators	0.8503	0.0399	0.0000
recent_roa	1.7600	0.1360	0.0000

Variable	Beta	Robust SE (HC1)	p-value
Intercept	5.5541	0.1468	0.0000
log_pledge	0.1266	0.0177	0.0000
margin	2.6853	0.4033	0.0000
fixed_cost_ada	0.0000	0.0000	0.0004
saturation	3.6243	0.3997	0.0000
log_delegators	0.7505	0.0556	0.0000
lifetime_roa	0.3743	0.0730	0.0000
recent_roa	1.5650	0.1424	0.0000